English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Software Engineering, 02.02.2026 Exam

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Justify every answer with clear, detailed calculations. Electronic devices (except ordinary calculators) and additional materials are prohibited.

Time allowed: 3 hours. Good luck.

Problem 1. (1 point) A website randomly displays one of m advertisements whenever the page loads. Let M be the number of advertisements not shown after n loads, and A the event that advertisement 1 has not been shown after n loads. Find $\mathbb{P}(M = 0)$, $\mathbb{E}[M]$, $\mathrm{Var}(M)$, $\mathbb{P}(M = 0|A^c)$ and $\mathbb{P}(A|M = 1)$.

Problem 2. (1 point) The number of comments on a popular influencer's latest post is $N \sim \mathrm{Po}(120)$. Each comment is positive with probability 30%, independently of the others. Let A be the number of positive comments.

1. Given exactly 100 comments, how many do you expect to be positive?
2. Given exactly 40 positive comments, what total number of comments do you expect?
3. What is the correlation between the positive and total comment counts?
4. Given exactly 120 comments, approximately what is the probability that the percentage of positive comments differs from 30% by more than 5 percentage points?

Problem 3. (1 point) In a fictional model, the National Statistical Institute represents quarterly economic growth by a random variable Θ (in percent). Before collecting data, growth is assumed to fluctuate around zero: $\Theta \sim \mathrm{N}(0, 4)$. A preliminary report based on incomplete data gives growth X. If true growth is $\Theta = \theta$, the preliminary estimate has conditional distribution $X \mid \Theta = \theta \sim \mathrm{N}(\theta, 1)$.

1. Given the preliminary estimate $X = x$, what true growth do you expect and with what uncertainty (variance)?
2. What is the probability of positive true growth if the preliminary report gives 1%?
3. What is the probability that reported growth is below true growth, given that true growth is positive?
4. Show that the reporting error $X - \Theta$ and true growth are independent.

Problem 4. (1 point) Let U_1, U_2 be independent $\mathrm{Unif}(0, 1)$ random variables. Put

$$X = U_1 - U_2, \qquad Y = |U_1 - U_2|, \qquad R = \frac{U_2}{U_1}.$$

Find f_X, $\mathbb{E}[X|U_1 > U_2]$, f_Y, $\mathbb{E}[Y]$ and the conditional distribution of R given $U_1 > U_2$.

Table entry for z is the area under the standard normal curve to the left of z.

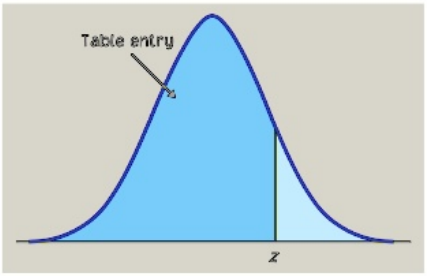


TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998